

Lab: Robotics and Embodied Active Inference

Learning Goal: Design and analyze Active Inference implementations for robotic systems.

Part 1: Discrete POMDP Design (pymdp)

Exercise: Design a discrete Active Inference agent for a simple grid navigation task:

Environment: 4×4 grid, agent starts at (0,0), goal at (3,3), obstacle at (2,2).

State space: 16 grid positions (4×4) **Observation space:** Visual observation of current position (with 10% noise) **Action space:** {up, down, left, right, stay}

POMDP specification:

Matrix	Dimensions	Description
A (observation)	16×16	Noisy identity (0.9 on diagonal, 0.1/15 elsewhere)
B (transition)	$16 \times 16 \times 5$	Deterministic grid transitions per action
C (preferences)	16×1	High value at (3,3), negative at (2,2), zero elsewhere
D (prior on start)	16×1	Peaked at (0,0)

Write the expected free energy $G(\pi)$ for a policy $\pi = [\text{right, right, down, down, down, right}]$:

- Pragmatic: Does it reach the goal?
- Epistemic: Does it reduce uncertainty about position?

Write your response here

Part 2: Continuous Control Design

Learning Goal: Design a continuous Active Inference controller.

Exercise: Design a controller for a robotic arm reaching task:

Generative model:

- States: joint angles $\theta = (\theta_1, \theta_2)$ and velocities $\dot{\theta} = (\dot{\theta}_1, \dot{\theta}_2)$
- Observations: end-effector position (x, y) via forward kinematics
- Desired state: end-effector at target position (x, y)

Free energy: $F = \frac{1}{2} \epsilon_s^T \Pi_s \epsilon_s + \frac{1}{2} \epsilon_p^T \Pi_p \epsilon_p$

Where:

- ϵ_s = sensory PE (observed position - predicted position)
- ϵ_p = prior PE (current position - desired position)
- Π_s, Π_p = precision matrices

Action generation: $a = -\partial F / \partial a$ — motor commands are the gradient of free energy with respect to action

Design the precision parameters Π_s and Π_p for:

1. Normal reaching (balanced vision and proprioception)
2. Vision occluded (low Π_s , rely on proprioception)
3. Novel environment (low Π_p , exploratory behavior)

Write your response here

Part 3: Framework Comparison

Learning Goal: Compare Active Inference software tools.

Exercise: Complete this comparison for implementing a T-maze navigation task:

Feature	pymdp (Python)	SPM (MATLAB)	RxInfer.jl (Julia)
State space	Discrete (POMDP)	Discrete (POMDP) + continuous (DCM)	Discrete + continuous
Inference method	Variational message passing	Variational Bayes	Reactive message passing
Speed	Moderate	Slower (MATLAB overhead)	Fast (Julia compilation)
Scalability	Small-medium state spaces	Small state spaces	Scales to larger models
Real-time capability	No (not designed for it)	No	Yes (reactive, streaming)
Hierarchical models	Manual construction	Built-in deep temporal	Factor graph composition
Community	Growing, open-source	Established, neuroimaging	Growing, probabilistic programming
Best for	Prototyping, education	Neuroimaging analysis	Real-time robotics, production

Which tool would you choose for: (a) a classroom demo? (b) an fMRI study? (c) a real-time robot controller?

Write your response here

Part 4: Human-Robot Interaction Design

Learning Goal: Design an HRI experiment using multi-agent Active Inference.

Exercise: Design a collaborative object sorting task:

1. **Robot's generative model of human:** Models human's preferences (C_{human}), attention (precision), and action tendencies
2. **Shared task model:** Both agents have models of the sorting rules and object locations
3. **Communication:** Robot infers human's sorting intent from observed actions; adjusts its behavior to complement (not duplicate) human actions
4. **Active Inference prediction:** Robot should anticipate human's next action and prepare complementary action

Describe: What observations does the robot use to model the human? How does it update its model in real-time? What does coordination look like when it succeeds vs. fails?

Write your response here

Part 5: Reflection

In 300 words, reflect: Active Inference robotics promises a unified perception-action framework, but current implementations are far from human-level dexterity. What are the main bottlenecks? Is the mathematical framework sound but the engineering immature? Or are there fundamental limitations to the approach?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	POMDP design	Discrete Active Inference
2	Controller design	Continuous generalized coordinates
3	Tool evaluation	Software framework comparison
4	Multi-agent robotics	Human-robot interaction
5	Critical evaluation	Approach limitations